Comprehensive 18-Week Syllabus Outline for Advanced Computer Science Paradigm

An intensive blueprint constructed optimized for university platforms targeting theory + practical.

Week 1: Foundations of Mobile Testing & Device Diversity

Topics:
- Definition of mobile testing vs desktop testing
- Visual layout and behavioral correctness across form factors
- Real-world failure modes and crash analysis in mobile environments

Objectives:
- Understand the fundamental differences between mobile and desktop application testing
- Analyze why multi-device testing is critical for modern software deployment

Tasks:
- Read introductory lecture notes on mobile testing basics
- Analyze a case study of a mobile application failure due to device mismatch

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Week 2: Android Fragmentation & Ecosystem Challenges

Topics:
- The concept of Android fragmentation and its impact on QA
- Managing thousands of unique device models and active OS versions
- Screen size variations (4 to 7 inches) and layout adaptation testing

Objectives:
- Identify the primary challenges associated with Android's open ecosystem
- Formulate a device selection matrix for representative test coverage

Tasks:
- Map out a target device matrix based on market share data
- Conduct a layout compatibility analysis on three different screen resolutions

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Week 3: iOS Testing Ecosystem & Platform-Specific Constraints

Topics:
- iOS testing challenges despite a more closed ecosystem
- Managing concurrent active iOS versions (iOS 15, 16, 17)
- Hardware constraints, Mac-only testing requirements, and App Store guidelines

Objectives:
- Compare and contrast testing workflows between Android and iOS platforms
- Navigate the strict submission rules and pre-release testing requirements of Apple

Tasks:
- Set up a simulated iOS environment on macOS
- Draft a pre-submission checklist matching App Store quality guidelines

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Week 4: Real-World Mobile Environmental Challenges

Topics:
- Network speed variability (5G, 3G, offline transitions)
- Battery consumption and memory footprint optimization
- Performance testing under resource-constrained hardware profiles

Objectives:
- Design test cases that simulate fluctuating network conditions
- Measure and analyze application battery drain and memory leaks

Tasks:
- Use network throttling tools to test app behavior under slow connections
- Profile an application's memory usage during intensive operations

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Week 5: Mobile Interruptions, Permissions, and Gestures Testing

Topics:
- Handling system interruptions (calls, texts, push notifications)
- Testing security permissions (camera, location, contacts) and negative flows
- Automating complex touch gestures (swipes, pinches, zooms, long-presses)

Objectives:
- Validate application recovery states after unexpected system interruptions
- Verify application stability when critical system permissions are denied

Tasks:
- Write test scenarios for permission denial recovery
- Manually execute and document complex gesture-based navigation flows

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Week 6: Introduction to Mobile Test Automation & Appium Philosophy

Topics:
- The role of automation in modern mobile release cycles
- Introduction to Appium as a cross-platform, open-source framework
- The philosophy of code reusability across Android and iOS

Objectives:
- Explain the benefits of open-source automation frameworks over proprietary tools
- Understand how Appium interacts with apps without modifying source code

Tasks:
- Research the history and maintenance of the Appium project
- Compare Appium with platform-native automation tools (Espresso, XCTest)

Week 7: Appium Architecture & The WebDriver Protocol

Topics:
- The three-tier Appium architecture: Test Script, Appium Server, and Mobile Device
- The role of the WebDriver Protocol in translating test commands
- Understanding client-server communication in test automation

Objectives:

- Deconstruct the step-by-step execution flow of an Appium command
- Explain the relationship between Selenium WebDriver and Appium

Tasks:
- Draw a sequence diagram of an Appium test command execution
- Inspect HTTP traffic between a test script and the Appium server

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Week 8: Appium Environment Setup & Dependency Management

Topics:
- Installing Node.js and managing global npm packages
- Installing the Appium server via command-line interface
- Configuring system environment variables (ANDROID_HOME, Java paths)

Objectives:
- Successfully configure a fully functional Appium testing environment
- Troubleshoot common environment path and dependency errors

Tasks:
- Install Node.js and Appium on a local workstation
- Configure system paths and verify the installation using terminal commands

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Week 9: Platform Drivers: UIAutomator2 & XCUITest Deep Dive

Topics:
- The role of platform-specific drivers in mobile automation
- Installing and configuring the UIAutomator2 driver for Android
- Installing and configuring the XCUITest driver for iOS

Objectives:
- Differentiate between Android and iOS automation drivers
- Install and initialize platform-specific drivers via the Appium CLI

Tasks:
- Install UIAutomator2 and run driver diagnostic checks
- Set up Android Studio or Xcode to manage virtual devices

Week 10: Scripting First Appium Tests & Element Locators

Topics:
- Writing basic Appium test scripts using Python/Java
- Defining Appium Options (app path, device name, platform version)
- Locating mobile elements using IDs, XPaths, and accessibility labels

Objectives:
- Write a complete script to launch a mobile application automatically
- Identify and interact with UI elements programmatically

Tasks:
- Write a script to launch a sample APK/APP file
- Locate a login button and programmatically trigger a click event

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Week 11: Real Devices vs. Emulators/Simulators in Test Pipelines

Topics:
- Pros and cons of virtual devices (emulators/simulators) vs real hardware
- Cost-efficiency and scalability of virtual environments
- Hardware limitations of emulators (camera, GPS, sensor testing)

Objectives:
- Determine when to use emulators versus real devices in a testing lifecycle
- Configure and run automated tests on both virtual and physical devices

Tasks:
- Run an automated test suite on an Android Emulator
- Connect a physical phone and execute the same test suite, comparing execution times

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Week 12: Advanced Appium Scripting & Test Execution Best Practices

Topics:
- Structuring clean test code and managing driver sessions
- Implementing explicit and implicit waits for dynamic UI loading
- Best practices: Emulators for daily development, real devices for pre-release

Objectives:
- Write robust, non-flaky automation scripts using proper synchronization
- Implement clean teardown procedures to prevent orphaned driver sessions

Tasks:
- Refactor a fragile test script to use explicit waits
- Implement a test suite that automatically closes sessions upon completion

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Week 13: Object-Oriented Systems Testing: Core Paradigms Review

Topics:
- Review of the 4 pillars of OOP: Encapsulation, Inheritance, Polymorphism, Abstraction
- How object-oriented design patterns affect software testability
- Unit testing principles for isolated classes and objects

Objectives:
- Analyze how object-oriented structures influence testing strategies
- Identify potential testing challenges unique to highly encapsulated code

Tasks:
- Review a codebase and identify examples of the 4 OOP pillars
- Write unit tests for a highly encapsulated class using public interfaces

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Week 14: Testing Inheritance: The Fragile Base Class & Overriding Pitfalls

Topics:
- The mechanics of inheritance and code reuse in testing
- The 'Fragile Base Class' problem: How parent changes break child classes
- Testing overridden methods and managing unexpected side effects

Objectives:
- Anticipate and identify regression bugs caused by modifications in parent classes
- Design test suites that validate overridden behaviors without breaking base assumptions

Tasks:
- Simulate a parent class modification and trace the failures in child classes
- Write tests specifically targeting overridden methods in a subclass

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Week 15: Strategies for Inheritance Testing: Isolation & Mocking

Topics:
- Isolating subclasses during unit testing
- Using mock objects to decouple parent-child dependencies
- Regression testing strategies for extended class hierarchies

Objectives:
- Isolate individual classes within an inheritance chain for targeted testing
- Apply mocking frameworks to simulate base class dependencies

Tasks:
- Create mock objects to isolate a subclass during unit testing
- Develop a regression test suite that automatically runs when a base class changes

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Week 16: Testing Polymorphism: Late Binding & Type-Specific Behaviors

Topics:
- The mechanics of polymorphism and dynamic binding at runtime
- Common pitfalls: Testing only a single type, late binding bugs
- Missing method runtime crashes in dynamically typed or poorly implemented subclasses

Objectives:
- Identify potential runtime failures caused by dynamic method binding
- Formulate test cases that cover all possible concrete implementations of an interface

Tasks:
- Write a test suite that executes a polymorphic method across multiple subclasses
- Simulate a missing method runtime crash and document the stack trace

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Week 17: Polymorphism Testing Strategies: Unit, Mocking, and Runtime Validation

Topics:
- The Golden Rule: Testing every class that uses a polymorphic method
- Writing unit tests with mock objects for polymorphic interfaces

- Validating runtime behavior and dynamic type resolution

Objectives:
- Apply the golden rule of polymorphism testing to a complex codebase
- Design integration tests that validate interactions between polymorphic classes

Tasks:
- Refactor a test suite to ensure every polymorphic subclass is tested independently
- Implement mock-based unit tests for an interface with multiple implementations

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Week 18: Integration Testing of OO Systems & Comprehensive Capstone Review

Topics:
- Transitioning from unit testing to integration testing in OO systems
- Testing how multiple classes and polymorphic interfaces interact together
- Course review: Synthesizing mobile testing, Appium automation, and OO testing

Objectives:
- Validate the end-to-end flow of an object-oriented mobile application
- Synthesize all course concepts into a cohesive software quality assurance strategy

Tasks:
- Perform integration testing on a multi-class system to find interaction bugs
- Present a final capstone project demonstrating automated mobile and OO testing

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